

Kidney Disease Prediction using ML, DL & XAI

1. Introduction

Chronic Kidney Disease (CKD) is a global health concern, affecting millions worldwide. Early detection and accurate prediction are critical for timely treatment and prevention of disease progression.

This project applies **Machine Learning (ML)**, **Deep Learning (DL)**, and **Explainable AI (XAI)** to:

- Predict kidney disease risk from clinical parameters and lab results.
- Compare model performance across ML and DL approaches.
- Interpret predictions to identify the most influential clinical factors.

Dataset: Chronic Kidney Disease dataset (400 patients × 25 features) with demographic, clinical, and blood test parameters.

2. Methodology

2.1 Exploratory Data Analysis (EDA)

- **Missing values:** Imputed using median (numerical) and mode (categorical).
- **Class imbalance:** Addressed using SMOTE (250 healthy vs. 150 CKD).
- **Key trends:** Hemoglobin and glucose levels strongly correlated with CKD status.
- **Visualization:** Correlation heatmaps highlighted strong associations among clinical parameters.

2.2 Preprocessing

- Encoding categorical variables (gender, specific gravity, etc.).
- Scaling features with StandardScaler.

- Target variable: CKD classification (0 = Healthy, 1 = CKD).
- Train-test split: 80/20 (stratified).

2.3 Models Implemented

Machine Learning Models: Logistic Regression, Decision Tree, Random Forest, SVM, KNN, XGBoost.

Deep Learning Models:

- MLP (Dense 64 → Dense 32 → Output).
- 1D CNN (Conv1D → Flatten → Dense).
- LSTM (64 units → Dense 32 → Output).
- Autoencoder + Classifier.

2.4 Evaluation Metrics

- Accuracy, Precision, Recall, F1-score.
- ROC-AUC.
- Confusion matrix analysis.

3. Results

3.1 Model Performance (Test Set)

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.92	0.92	0.92	0.92	0.97
Decision Tree	0.94	0.94	0.94	0.94	0.94
Random Forest	0.96	0.96	0.96	0.96	0.99
SVM	0.93	0.93	0.93	0.93	0.98
KNN	0.91	0.91	0.91	0.91	0.96
XGBoost	0.95	0.95	0.95	0.95	0.99
MLP	0.94	0.94	0.94	0.94	0.98

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
CNN-1D	0.92	0.92	0.92	0.92	0.96
LSTM	0.91	0.91	0.91	0.91	0.95
Autoencoder+Classifier	0.93	0.93	0.93	0.93	0.97

Key Observations:

- **Tree-based models (Random Forest, XGBoost)** consistently outperformed others.
- **DL models** performed competitively but required more computation and fine-tuning.
- **Random Forest** offered the best balance of accuracy and efficiency.

4. Explainable AI (XAI) Insights

- **Feature Importance (Tree Models):** Hemoglobin most critical, followed by glucose, age, serum creatinine, and blood pressure.
- **SHAP Values:** Lower hemoglobin levels strongly linked with CKD. Local explanations revealed patient-specific risk factors.
- **PDP & ICE Plots:** Confirmed hemoglobin and glucose as key risk drivers, with non-linear age influence.
- **Neural Network Interpretability:** Integrated Gradients aligned with tree models, highlighting hemoglobin and glucose.

5. Comparative Analysis

Aspect	ML (Tree-Based)	DL (Neural Nets)
Accuracy	High (RF: 96%)	Competitive (MLP: 94%)
Training Time	Seconds	Minutes (with GPU)
Interpretability	Strong (FI, SHAP, PDP)	Limited (needs XAI tools)
Clinical Trust	High	Moderate

Aspect	ML (Tree-Based)	DL (Neural Nets)
Adoption	Easier	Requires further validation

Takeaway: Random Forest balances performance and interpretability, making it suitable for clinical use.

6. Clinical Implications

- **Early Screening:** Focus on patients with declining hemoglobin levels.
 - **Risk Stratification:** Identify high-risk groups using model insights.
 - **Personalized Medicine:** Use patient-specific explanations for treatment planning.
 - **Preventive Care:** Monitor key parameters (Hb, glucose, BP) in routine checkups.
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7. Limitations & Future Work

- **Dataset Size:** Small (400 patients); needs larger, multi-center validation.
 - **Feature Expansion:** Genetic and lifestyle factors could improve predictions.
 - **Temporal Data:** Longitudinal tracking may aid in progression modeling.
 - **Deployment:** Build real-time clinical decision support systems with XAI integration.
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8. Conclusion

This study confirms that **ML and DL models can predict CKD effectively**, with **Random Forest achieving the best performance (96% accuracy, 0.99 ROC-AUC)**.

XAI methods reinforce medical relevance by identifying **hemoglobin, glucose, and age** as dominant risk factors.

The integration of ML, DL, and XAI fosters **trust, transparency, and actionable clinical insights**, paving the way for early detection and better patient care.
